

2025 Integrated Work Study Programme (IWSP) at OMRON as a Robotics Application Engineer

ABOUT OMRON

OMRON Electronics Pte Ltd Singapore (OEPSG) is an **Industrial Automation** partner that creates, sells and services fully integrated automation solutions that include sensing, control, safety, vision, motion, **robotics** and more.



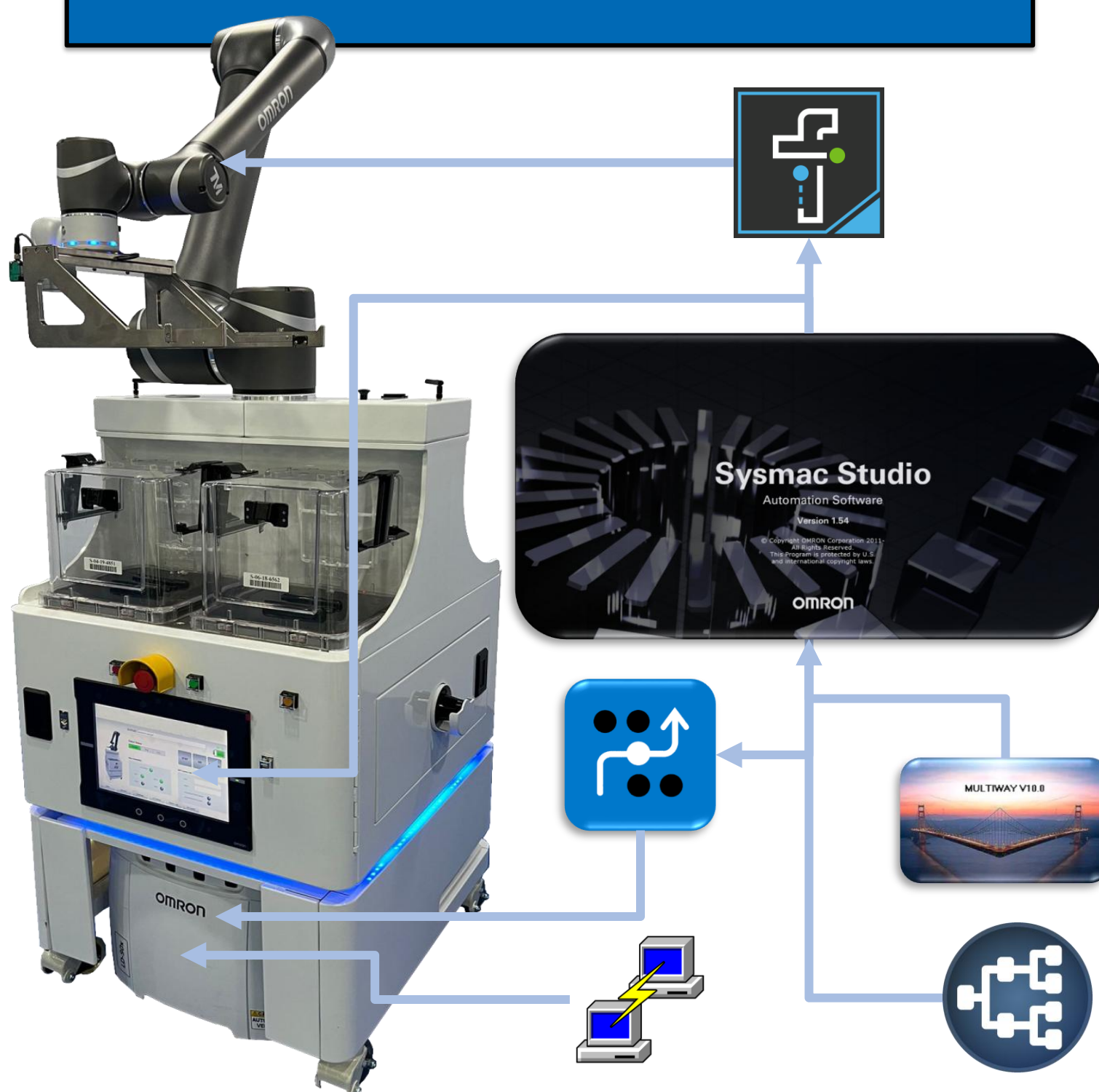
OMRON SG SG60 Photoshoot

JOB SCOPE

- Provided on-site customer **support**.
- Monitored **deployed robotics solutions**.
- Triaged customer-reported incidents, reproduced issues, and **verified fixes**.
- Performed in-lab **acceptance testing**.
- **Documented** findings and updated status.

INTERNSHIP LEARNING JOURNEY

PROJECTS & TASKS



Mobile Manipulator (MoMa) robot introduces **Industrial Automation** of wafer cassette handling, replacing a previously labour-intensive process.

Co-authored **decision-making & navigation stack** and **mapped** real-world environment for **traversal** and **docking**.

Taught Pick-and-Place routines at defined pickup/dropoff locations and performed end-to-end **integration** with error-recovery sequences **implemented**

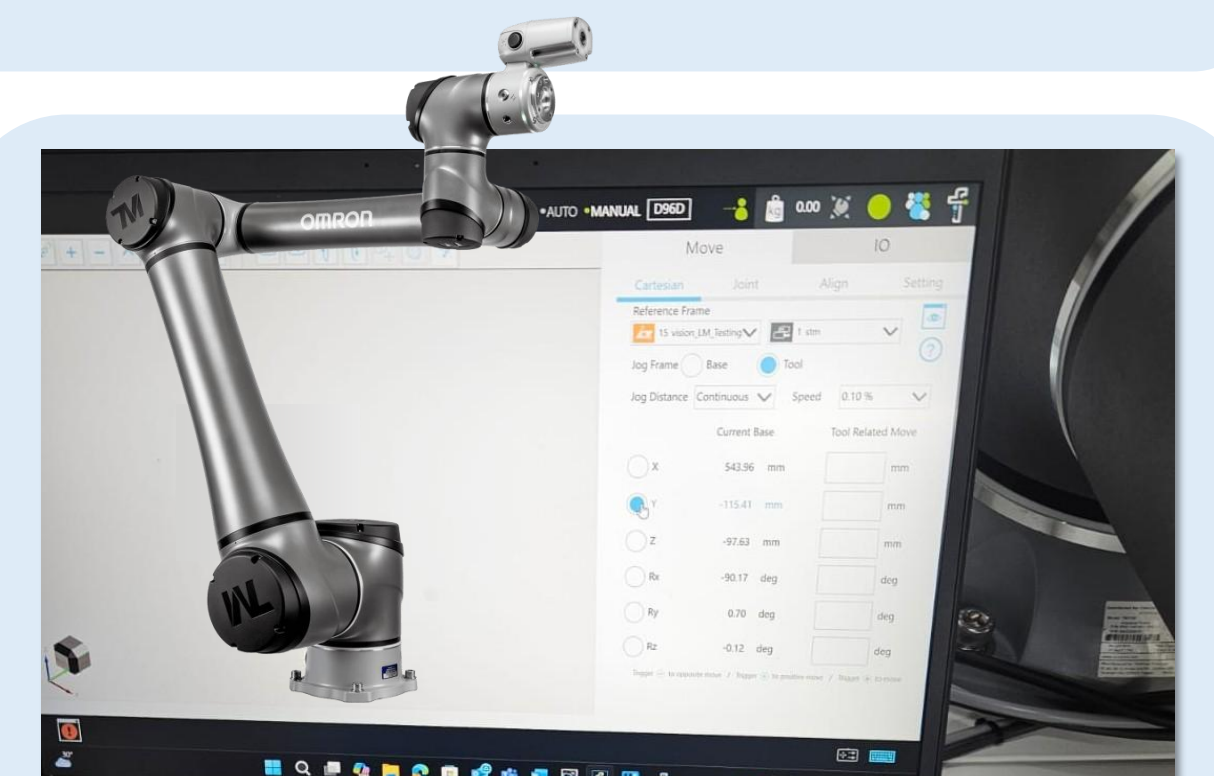
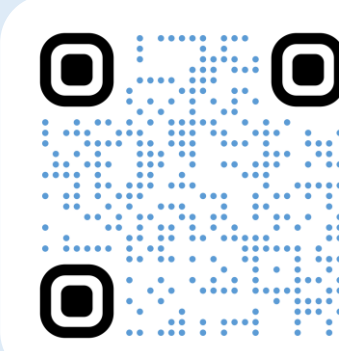
Developed a generic CPP Inference Application Programming Interface (API) for Open Neural Network Exchange (ONNX) formats **integrated** with OMRON vision controllers built on ONNX Runtime (ORT) for **optimization** of inference speed.



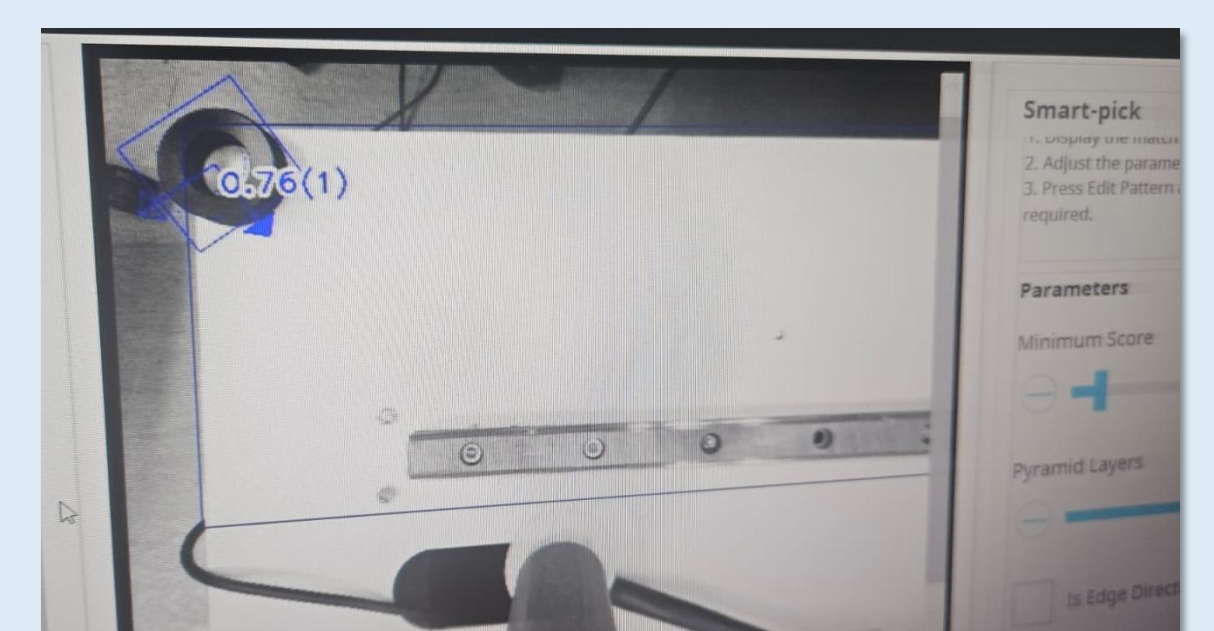
Wrote a script for an automated **2D imaging pipeline** of OMRON SENTECH Liquid Lens Industrial Cameras using OpenCV for **processing** and StApiPy for **device control**.



Created **Digital Twin** simulation scenarios in **Mobile Planner software** using Virtual Fleet Manager (VFM) for **path-planning**, **traffic management** and in-lab **validation** prior to site **deployment**.



Built Pick-and-Place routines in **TMFlow software** on OMRON TM Robotic arm and delivered stakeholder **Proof-Of-Concept (POC) demonstration** videos for picking circular, hollow-centre objects.



EXPERIENCE

Working alongside engineers gave me exposure to *professional industry experience* contrasting beyond being an *academic* in a classroom.

- Adopting an **open, critical** mind by asking engineers questions to clarify any assumptions early.
- **Relearning** to apply what was taught in an unfamiliar and new environment and **translating theory into application**.
- The **learning never stops**.



OMRON International Sushi Day



OMRON Team Bonding Day